

Geometry of Splats

Chizat Bach 2018: *conditional* global optimization for 2-layer ReLU/Sigmoid nets via WGF, for,

$$\mathcal{F}(\mu) = R \left(\int \Phi d\mu \right) + \int V d\mu \quad \mu \in \mathcal{P}_2(\Omega), \Omega \subset \mathbb{R}^d$$

Here: $\Phi(v, W, b) = v (Wx + b)_+$, risk $R(\cdot)$, regularizer $V(\cdot)$

Assumptions:

- R is convex, differentiable, Lipschitz on bdd sets, bdd on sublevel sets.
- V is semiconvex ($V(\theta) + \lambda |\theta|^2$ is convex for some $\lambda > 0$)
- V, Φ are 2-homogeneous functions. (satisfied by ReLU + Ridge reg.)
- μ_0 support satisfies a **separation condition**.

Geometry of Splats

Definition (Chizat, Bach 2018 – informal)

Fix $\mu_0 \in \mathcal{P}(\Omega)$, $\Omega \subseteq \mathbb{R}^d$, and for $\mu \in \mathcal{P}(\Omega)$ set:

$$\mathcal{F}(\mu) = R \left(\int \Phi d\mu \right) + \int V d\mu$$

Assume (...) hold. Then, whenever the WGF $(\mu_t)_{t \geq 0} \rightarrow \mu_\infty$ in W_2 , also

$$\lim_{t \rightarrow \infty} \mathcal{F}(\mu_t) = \min_{\mu \in \mathcal{M}_+} \mathcal{F}(\mu)$$

Further: if $(\mu_{m,t})_{m \geq 1, t > 0}$ is a sequence of gradient flows $\mu_{m,0} \rightarrow \mu_0$ in W_2 , then limits exchangeable:

$$\lim_{m, t \rightarrow \infty} \mathcal{F}(\mu_{m,t}) = \min_{\mu \in \mathcal{M}_+} \mathcal{F}(\mu)$$